

- 2 A curve has equation

$$(x+1)y + y^2 = 2.$$

- (a) Show that $\frac{dy}{dx} = -\frac{2}{3}$ at the point $(0, -2)$. [3]

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- (b) Find the value of $\frac{d^2y}{dx^2}$ at the point $(0, -2)$. [4]

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